THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A smooth muscle cell has: 1

(a) many nuclei **(b)** striations
(c) a single nucleus and spindle shape **(d)** branching

2. The binomial name 'Canis lupus' and 'Canis familiaris' show that the two: 1

(a) belong to different families
(b) belong to the same species
(c) belong to the same genus but different species
(d) belong to different genera

3. The grafting is successful when the: 1

(a) cambia of stock and scion are in contact **(b)** xylems are far apart
(c) roots are removed **(d)** leaves are cut

4. The stratosphere is warmer at the top due to the: 1

(a) absorption of UV by ozone **(b)** volcanoes
(c) oceans **(d)** ground heating

5. An experimental result is 4.8 and the model gives 5.0. The first step in an honest investigation is to: 1

(a) check the experiment and the assumptions in the model
(b) ignore the model
(c) change the 4.8 to 5.0
(d) drop the experiment

6. A student mixes sand and powdered sugar. The best method of separation is: 1

(a) magnet **(b)** winnowing
(c) dissolve in water, filter, evaporate **(d)** sublimation

7. The percentage of carbon in CO_2 is about (C = 12, O = 16): 1

(a) 72.7 per cent **(b)** 27.3 per cent **(c)** 12 per cent **(d)** 50 per cent

8. Which sub-atomic particle has a negligible mass compared with the other two? 1

(a) electron **(b)** proton
(c) neutron **(d)** none of them has a negligible mass

9. A body covers the first half of its journey at 20 m/s and the second half at 30 m/s. The average speed is: 1

(a) 24 m/s **(b)** 25 m/s **(c)** 50 m/s **(d)** 26 m/s

10. Assertion (A): A heavier object is harder to set in motion than a lighter one. Reason (R): A heavier object has more inertia. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): A body thrown up has maximum potential energy at the highest point. Reason (R): The kinetic energy is maximum there. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): A louder sound has a greater frequency. Reason (R): Loudness depends on the amplitude only. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Assertion (A): Mitochondria are called the powerhouse of the cell. Reason (R): They release energy from food. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. The impulse in a neuron travels from: 1

(a) axon to dendrite **(b)** axon to cell body
(c) dendrite to axon **(d)** cell body to dendrite

15. A cell with a cell wall, chloroplast and nucleus belongs to: 1

(a) Monera **(b)** Plantae **(c)** Fungi **(d)** Animalia

16. Seeds dispersed by animals have: 1

(a) no coat **(b)** hooks or fleshy fruit
(c) air bladders **(d)** wings

17. Assertion (A): The natural greenhouse effect is essential for life on Earth. Reason (R): Without it the Earth would be too cold to support life. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. Assertion (A): A measurement can never be exactly correct. Reason (R): Every instrument has a least count and every observation has some uncertainty. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. Assertion (A): The solute in a copper sulphate solution cannot be separated by filtration. Reason (R): The solute particles are much smaller than the pores of the filter paper. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. Assertion (A): Carbon tetrachloride has a lower melting point than sodium chloride. Reason (R): It is a covalent compound with weak forces between molecules. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

21. Assertion (A): Rutherford's model could not explain the stability of the atom. Reason (R): According to classical physics an accelerating electron radiates energy and would spiral into the nucleus. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

22. Assertion (A): A ball thrown vertically up has zero acceleration at the highest point. Reason (R): Its velocity is zero at the highest point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

23. A balloon moving upward at constant velocity releases a stone. Just after release, the stone's acceleration is: 1

(a) 10 m/s squared upward **(b)** zero
(c) 20 m/s squared **(d)** 10 m/s squared downward

24. The ratio of KE of two bodies of masses 1 kg and 4 kg with equal momentum is: 1

(a) 2 : 1 **(b)** 4 : 1 **(c)** 1 : 1 **(d)** 1 : 4

25. A person hears the echo of a gun shot 1 s after the shot and a second echo 3 s after. The ratio of the distances of the two reflectors is: 1

(a) 3 : 1 **(b)** 1 : 1 **(c)** 1 : 3 **(d)** 1 : 2

26. A cell with 20 chromosomes divides by mitosis and its daughter cells again divide by meiosis. The number of chromosomes in the final cells is: 1

(a) 10 **(b)** 40 **(c)** 5 **(d)** 20

27. Which of these stains is used for a section of a stem to show the xylem? 1

(a) eosin **(b)** iodine **(c)** safranin **(d)** methylene blue

28. Bats and birds both fly but are placed in different classes because bats: 1

(a) have scales **(b)** have feathers
(c) lay eggs **(d)** have hair and mammary glands

29. The male gland that supplies fluid to the sperm is the: 1

(a) adrenal **(b)** thyroid
(c) pituitary **(d)** prostate and seminal vesicles

30. A tree absorbs 22 kg of CO2 per year. The number of trees needed to offset the 2200 kg emission of a family is: 1

(a) 1000 **(b)** 22 **(c)** 100 **(d)** 10

31. A cube of side 2.0 cm (+/- 0.1 cm). The percentage error in the volume is about: 1

(a) 15 per cent **(b)** 1.5 per cent **(c)** 10 per cent **(d)** 5 per cent

32. In a chromatogram of an ink with R_f values 0.3, 0.6, 0.9, the component most soluble in the solvent is: 1

(a) the one with R_f 0.9 **(b)** the one with R_f 0.6
(c) the one with R_f 0.3 **(d)** all equal

33. The formula unit mass of the compound $\mathrm{Ca_3(PO_4)_2}$ is (Ca = 40, P = 31, O = 16): 1

(a) 262 u **(b)** 230 u **(c)** 310 u **(d)** 278 u

34. An element has mass number 31 and 16 neutrons. The number of valence electrons is: 1

(a) 3 **(b)** 15 **(c)** 8 **(d)** 5

35. The ratio of distances covered in the 1st, 2nd and 3rd seconds from rest with uniform acceleration is: 1

(a) 1 : 1 : 1 **(b)** 1 : 2 : 3 **(c)** 1 : 4 : 9 **(d)** 1 : 3 : 5

36. A car and a truck, both with the same momentum, are brought to rest in the same time. The retarding force is: 1

(a) the same for both **(b)** greater for the truck
(c) greater for the car **(d)** zero

37. A mass of 2 kg is raised by 3 m with an acceleration of 2 m/s squared from rest (g = 10). The work done by the lifting force is: 1

(a) 72 J **(b)** 12 J **(c)** 60 J **(d)** 36 J

38. A stone is dropped into a well and the splash is heard 2.1 s later. If the speed of sound is 340 m/s and g = 10, the depth is closest to: 1

(a) 25 m **(b)** 20 m **(c)** 15 m **(d)** 35 m

39. A cell biologist adds a poison that blocks the mitochondria. The process first affected is: 1

(a) energy release **(b)** osmosis
(c) protein synthesis **(d)** diffusion

40. Assertion (A): Xylem vessels are dead cells yet efficient in conducting water. Reason (R): Their hollow lignified walls form continuous tubes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. **Assertion (A):** Sound waves in air are longitudinal waves. 1
Reason (R): The speed of sound is greater in solids than in gases.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Plant cells have a cell wall, but animal cells do not. 1
Reason (R): The plant cell wall is made mainly of cellulose.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Phloem transports food from the leaves to other parts of the plant. 1
Reason (R): Xylem is composed mainly of dead cells.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The scientific name of the human being is *Homo sapiens*. 1
Reason (R): In binomial nomenclature the first word is the genus (beginning with a capital letter) and the second word is the species (beginning with a small letter).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Yeast reproduces by budding. 1
Reason (R): *Amoeba* reproduces by binary fission.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Carbon dioxide is called a greenhouse gas. 1
Reason (R): Nitrogen makes up about 78 per cent of the atmosphere.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** A model can be useful in science even though it is simplified. 1
Reason (R): A scientific model is an exact copy of the real object or system.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** A suspension is a heterogeneous mixture. 1
Reason (R): The particles of a suspension do not settle down when the mixture is left undisturbed.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The chemical formula of aluminium oxide is Al_2O_3. 1
Reason (R): Aluminium has valency 2 and oxygen has valency 3.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The mass number of an atom equals the number of protons plus the number of neutrons. 1
Reason (R): Electrons contribute most of the mass of an atom.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**

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